

Conjecture 4

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id 2603.02786__01 — generated 2026-09-17

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result would_publish: true and returned proved or disproved. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim below as unverified.

Summary and provenance

Catalog id	2603.02786__01
Catalog entry	Conjecture 4
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Packing arithmetic progressions
Model verdict	proved (confidence: high)
Model's one-line claim	The conjectured asymptotic follows from an exact packing lemma for arithmetic-progression blocks of differences and the dense-subset Green–Tao theorem.
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2603.02786__01/output.md
Independent review	None
Model's caveats	Uses the established relative-density Green–Tao theorem; novelty has not been checked against the full source paper.

Problem statement

No extracted statement was stored in the artifact metadata.

Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

Proof of the conjecture

The restriction in the previous attempt—that nonconsecutive convex hulls must be disjoint—can be bypassed. For suitable arithmetic-progression blocks of prime differences, one can coordinate **every pair** of progressions, even when many convex hulls overlap. The dense-subset Green–Tao theorem supplies a partition of almost all primes into such blocks.

All shifts below are integers. An integer interval of length L contains L consecutive integers; using diameter instead changes the packing parameter by one.

Write

$$A_d(b) = \{b, b+d, \dots, b+(n-1)d\}, \quad S(n) = \sum_{p \leq n} p(n-p).$$

We prove the slightly more informative statement

$$M_{\mathbb{P}(n)}(n) = S(n) + o\left(\frac{n^3}{\log n}\right).$$

The prime number theorem then gives the conjectured constant $1/6$.

1. An exact pairwise avoidance criterion

Lemma 1 Let p, q be coprime positive integers. Suppose

$$b' - b = Ap + Bq$$

for integers A, B satisfying

$$A \geq n - q, \quad B \geq 1.$$

Then $A_p(b)$ and $A_q(b')$ are disjoint.

Proof An intersection would give

$$b + up = b' + vq$$

for some $0 \leq u, v < n$, and hence

$$(u - A)p = (v + B)q.$$

Since the right-hand side is positive and $\gcd(p, q) = 1$, the integer $u - A$ is a positive multiple of q . Consequently,

$$u \geq A + q \geq n,$$

a contradiction. $\square$

The next lemma arranges for this criterion to hold simultaneously for every pair in a block.

2. Packing an arithmetic-progression block of differences

Lemma 2 Fix $k \geq 2$, and put

$$Q = \text{lcm}(1, 2, \dots, k-1).$$

Suppose

$$d_i = a + ih, \quad 0 \leq i < k,$$

are pairwise coprime positive integers satisfying

$$d_0 < d_1 < \dots < d_{k-1} \leq n, \quad Q \mid h.$$

Then their n -term progressions can be packed into an integer interval of length at most

$$\sum_{i=0}^{k-1} d_i(n - d_i) + n^2 + 2Qkn.$$

Proof Choose the integer n_* satisfying

$$n \leq n_* \leq n + Q - 1, \quad n_* \equiv a \pmod{Q}.$$

Thus every $n_* - d_i$ is a nonnegative multiple of Q .

Set $b_0 = 0$, and recursively define

$$b_{i+1} = b_i + d_i(n_* - d_{i+1}) + Qd_{i+1} \quad (0 \leq i < k-1).$$

We claim that all sets $A_{d_i}(b_i)$ are pairwise disjoint.

Fix $i < j$, and write

$$m = j - i, \quad p = d_i, \quad q = d_j.$$

For $0 \leq t < m$, the arithmetic-progression structure gives

$$d_{i+t} = \frac{m-t}{m}p + \frac{t}{m}q$$

and

$$d_{i+t+1} = \frac{m-t-1}{m}p + \frac{t+1}{m}q.$$

Therefore

$$\begin{aligned} d_{i+t}(n_* - d_{i+t+1}) &= \frac{(m-t)(n_* - d_{i+t+1})}{m}p \\ &\quad + \frac{t(n_* - d_{i+t+1})}{m}q, \end{aligned}$$

while

$$Qd_{i+t+1} = \frac{Q(m-t-1)}{m}p + \frac{Q(t+1)}{m}q.$$

Every coefficient displayed here is a nonnegative integer. Indeed, $m \leq k-1$, so $m \mid Q$, and $Q \mid n_* - d_{i+t+1}$.

Summing these identities over $t = 0, \dots, m-1$, we obtain

$$b_j - b_i = Ap + Bq$$

with integer coefficients. The first term of the first sum contributes $(n_* - d_{i+1})p$, so

$$A \geq n_* - d_{i+1} \geq n - q.$$

The last term of the second sum contributes Qq , so

$$B \geq Q \geq 1.$$

Lemma 1 now proves disjointness for this pair. Since $i < j$ was arbitrary, all pairs are covered.

It remains to bound the containing interval. The b_i increase strictly, and the d_i increase, so the largest endpoint belongs to the last progression. Its containing interval has length

$$\Lambda = \sum_{i=0}^{k-2} [d_i(n_* - d_{i+1}) + Qd_{i+1}] + (n-1)d_{k-1} + 1.$$

For $i < k-1$, using $d_{i+1} \geq d_i$,

$$d_i(n_* - d_{i+1}) \leq d_i(n - d_i) + (n_* - n)d_i.$$

Also,

$$(n-1)d_{k-1} + 1 = d_{k-1}(n - d_{k-1}) + d_{k-1}^2 - d_{k-1} + 1 \leq d_{k-1}(n - d_{k-1}) + n^2.$$

Because $n_* - n < Q$ and all $d_i \leq n$, these inequalities give

$$\Lambda \leq \sum_{i=0}^{k-1} d_i(n - d_i) + n^2 + 2Qkn.$$

This proves the lemma. $\square$

The crucial feature is the cost: a block of k differences incurs only one n^2 -sized boundary overhead, plus $O_k(n)$. No separation condition on nonconsecutive convex hulls is imposed.

3. Almost all primes can be partitioned into suitable blocks

The non-elementary input is the following established form of the Green–Tao theorem.

Dense-subset Green–Tao theorem For every fixed $k \geq 3$ and $\delta > 0$, all sufficiently large N have the following property: every set

$$E \subseteq \mathbb{P}(N), \quad |E| \geq \delta \pi(N),$$

contains a nonconstant k -term arithmetic progression.

The dense-subset form matters here: the mere existence of arbitrarily long prime arithmetic progressions would not suffice.

Lemma 3 For any fixed positive integers $k \geq 3$ and Q , all but

$$o_{k,Q}(\pi(n))$$

primes up to n can be partitioned into k -term arithmetic progressions whose common differences are divisible by Q .

Proof Take a maximal collection of disjoint k -term arithmetic progressions of primes up to n , each with common difference divisible by Q . Let R_n be the unused primes.

If $|R_n| \geq \delta \pi(n)$, some residue class modulo Q contains at least

$$\frac{\delta}{Q} \pi(n)$$

members of R_n . For sufficiently large n , the dense-subset Green–Tao theorem supplies a k -term arithmetic progression within that residue class. Its common difference is divisible by Q , contradicting maximality.

This works for every fixed $\delta > 0$, proving the claim. $\square$

4. The asymptotically optimal upper bound

Fix $k \geq 3$, and let

$$Q = \text{lcm}(1, \dots, k-1).$$

Apply Lemma 3. Let t_n be the number of resulting blocks and u_n the number of unused primes. Then

$$kt_n + u_n = \pi(n), \quad u_n = o_k(\pi(n)).$$

Each block consists of distinct primes, hence its differences are pairwise coprime. Lemma 2 applies.

Pack the blocks into consecutive, disjoint integer intervals. Pack each unused prime progression into its own interval. For a singleton prime $p \leq n$, the required length satisfies

$$(n-1)p + 1 = p(n-p) + (p^2 - p + 1) \leq p(n-p) + n^2.$$

Adding all block and singleton costs gives

$$\begin{aligned} M_{\mathbb{P}(n)}(n) &\leq S(n) + (t_n + u_n)n^2 + 2Qkn t_n \\ &\leq S(n) + \frac{n^2 \pi(n)}{k} + u_n n^2 + 2Qn \pi(n). \end{aligned}$$

Here k and Q are fixed. Consequently,

$$\limsup_{n \rightarrow \infty} \frac{M_{\mathbb{P}(n)}(n) - S(n)}{n^2 \pi(n)} \leq \frac{1}{k}.$$

Since this holds for every fixed $k \geq 3$,

$$\limsup_{n \rightarrow \infty} \frac{M_{\mathbb{P}(n)}(n) - S(n)}{n^2 \pi(n)} \leq 0.$$

The order of limits is important: first $n \rightarrow \infty$ with k fixed, and only then $k \rightarrow \infty$. Thus the rapid growth of Q , and the lack of a useful quantitative Green–Tao bound, cause no difficulty.

5. Matching lower bound

For completeness, here is an independent proof of the required lower bound.

Consider any packing, and restrict it to the primes $p < n$. Order these progressions by their first terms:

$$a_1 < a_2 < \dots < a_s,$$

with corresponding differences $p_1, \dots, p_s$.

For consecutive progressions, put $p = p_i, q = p_{i+1}$. If

$$a_{i+1} - a_i \leq (n-1)p - pq,$$

then the integer interval

$$[a_{i+1}, a_{i+1} + pq - 1]$$

lies inside both convex hulls. For the second hull, this uses

$$(n-1)q \geq pq,$$

since $p \leq n-1$. The Chinese remainder theorem supplies a point in this interval congruent to $a_i \pmod{p}$ and to $a_{i+1} \pmod{q}$. Such a point belongs to both progressions, a contradiction.

Hence

$$a_{i+1} - a_i > (n-1)p_i - p_i p_{i+1}.$$

Summing and adding the span of the last progression yields the weaker, sufficient inequality

$$M_{\mathbb{P}(n)}(n) \geq (n-1) \sum_{i=1}^s p_i - \sum_{i=1}^{s-1} p_i p_{i+1}.$$

Using $2xy \leq x^2 + y^2$,

$$\sum_{i=1}^{s-1} p_i p_{i+1} \leq \sum_{i=1}^s p_i^2.$$

It follows that

$$M_{\mathbb{P}(n)}(n) \geq \sum_{p < n} p(n-p) - \sum_{p < n} p.$$

If n itself is prime, its contribution to $S(n)$ is zero. Therefore

$$M_{\mathbb{P}(n)}(n) \geq S(n) - n\pi(n).$$

In particular,

$$\liminf_{n \rightarrow \infty} \frac{M_{\mathbb{P}(n)}(n) - S(n)}{n^2 \pi(n)} \geq 0.$$

Together with the upper bound, this proves

$$M_{\mathbb{P}(n)}(n) = S(n) + o(n^2 \pi(n)).$$

6. Evaluating the main term

By the prime number theorem and partial summation,

$$\sum_{p \leq n} p = \left(\frac{1}{2} + o(1)\right) \frac{n^2}{\log n}, \quad \sum_{p \leq n} p^2 = \left(\frac{1}{3} + o(1)\right) \frac{n^3}{\log n}.$$

Thus

$$\begin{aligned} S(n) &= n \sum_{p \leq n} p - \sum_{p \leq n} p^2 \\ &= \left(\frac{1}{6} + o(1)\right) \frac{n^3}{\log n}. \end{aligned}$$

Also $n^2 \pi(n) \sim n^3 / \log n$. Consequently,

$$M_{\mathbb{P}(n)}(n) \sim \frac{1}{6} \frac{n^3}{\log n}.$$

All collision cases, including nonconsecutive progressions and primes above $n/2$, are covered by Lemma 2. The proof uses no unproved prime-pattern hypothesis; its only deep input is the established dense-subset Green–Tao theorem.